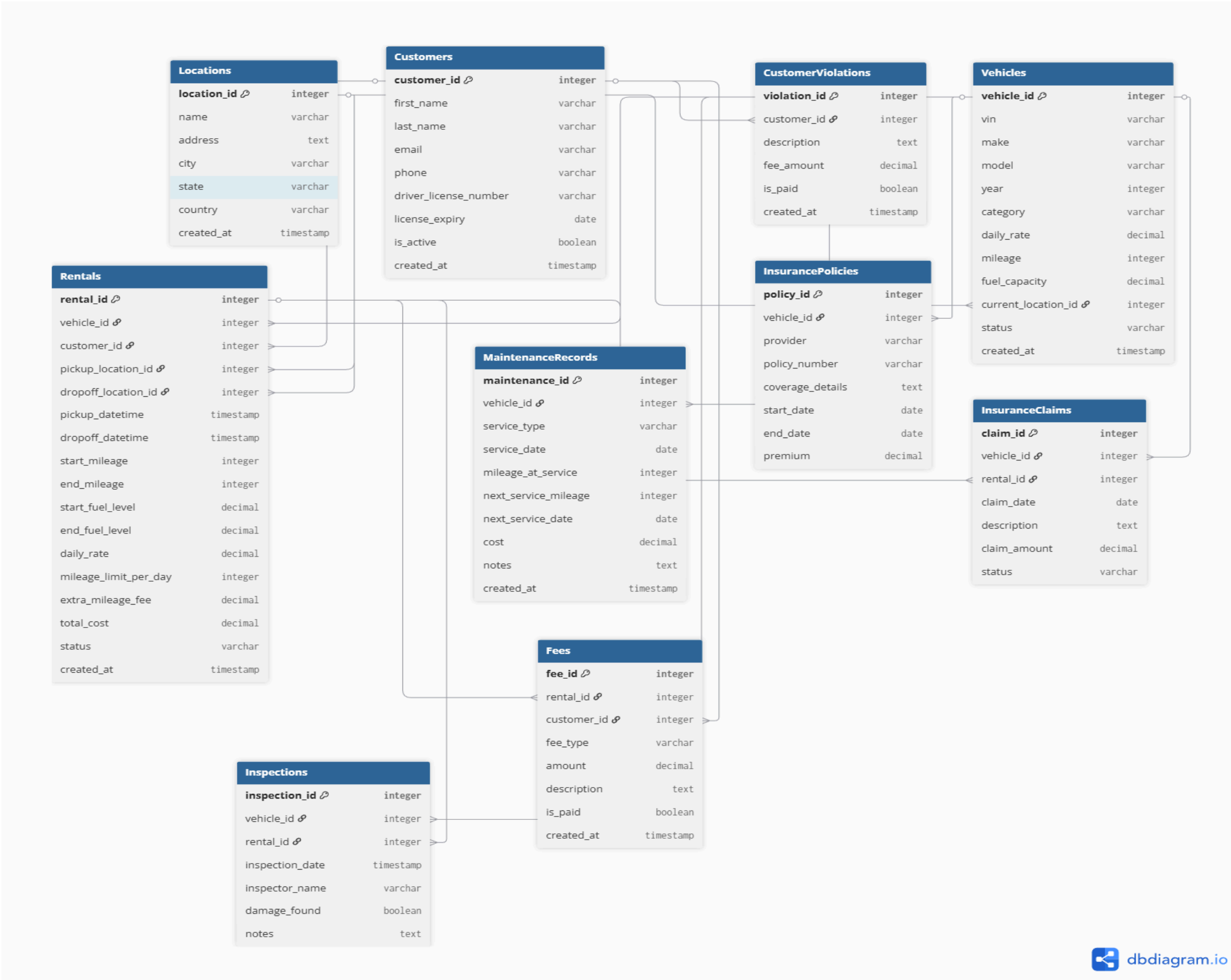

Justification for Normalization Decisions

Each table stores one main type of data.

- Vehicle data is stored in Vehicles, not inside Rentals.
- Customer data is stored in Customers, not repeated in every rental.
- Maintenance, insurance, inspections, and fees are in separate tables because they happen at different times and may occur many times for one vehicle.

Examples:

- A vehicle can have many maintenance records → stored in MaintenanceRecords
- A customer can have many rentals → stored in Rentals
- A rental can have many fees → stored in Fees

This avoids:

- repeating the same data

- update errors
- data inconsistency

The design follows:

- 1NF: no repeating fields
- 2NF: all data depends on the primary key
- 3NF: no unnecessary duplicated information

Identification of Critical Indexes (Simple)

Indexes help the system find data faster, especially for rentals and searches.

Important indexes

Vehicle availability

```
CREATE INDEX idx_vehicle_status_location
ON Vehicles(status, current_location_id);
```

Rental lookups

```
CREATE INDEX idx_rentals_vehicle
ON Rentals(vehicle_id);
```

Customer validation

```
CREATE INDEX idx_customer_license
ON Customers(license_expiry);

CREATE INDEX idx_customer_violations
ON CustomerViolations(customer_id, is_paid);
```

Maintenance checks

```
CREATE INDEX idx_maintenance_vehicle
ON MaintenanceRecords(vehicle_id);

CREATE INDEX idx_maintenance_next
ON MaintenanceRecords(next_service_date);
```

Location tracking

```
CREATE INDEX idx_vehicle_location
```

ON Vehicles(current_location_id);

These indexes are critical because they support:

- checking if a vehicle is available
- verifying customer eligibility
- tracking maintenance schedules
- finding rentals quickly
- managing vehicles across locations